The effect of plough agriculture on gender roles: A machine learning approach

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1 Introduction

The origins of global differences in gender roles and female labor force participation (FLFP) have long puzzled economists and sociologists. A prominent historical explanation is the "Boserup hypothesis," which posits that the historical adoption of plough agriculture generated an enduring gender division of labor (Boserup, 1970). Unlike shifting cultivation, which relies on the hoe and allows for high female participation, plough agriculture requires significant upper-body strength. This technological shift historically relegated women to the domestic sphere, generating deep-rooted cultural norms regarding gender roles that persist long after societies have industrialized.

In their paper, Alesina, Giuliano, and Nunn (2013) empirically tested this hypothesis, finding robust evidence that societies with a historical legacy of plough agriculture exhibit significantly lower FLFP and less equal gender norms today. Using traditional Ordinary Least Squares (OLS) and Instrumental Variable (IV) strategies, they established a strong negative correlation between historical plough use and contemporary gender equality. However, traditional econometric approaches to robustness—such as adding varying sets of control variables to linear models—can be somewhat ad-hoc and susceptible to omitted variable bias or functional form misspecification (Baiardi & Naghi, 2024).

To rigorously test the stability of these foundational findings, Baiardi and Naghi (2024) revisit the plough hypothesis utilizing recent advancements in Causal Machine Learning (CML), specifically the Double Machine Learning (DML) framework pioneered by Chernozhukov et al. (2018). Machine learning methods offer a systematic, data-driven approach to selecting model specifications, allowing researchers to control for a high-dimensional set of potential confounders (e.g., highly detailed geographic, historical, and economic variables) and their complex, non-linear interactions.

In this report, we synthesize the methodology and findings of Baiardi and Naghi (2024), highlighting the integration of machine learning within traditional causal inference frameworks. By replicating their conceptual approach, including a discussion of

the implementation in R, we demonstrate how CML serves as a powerful tool for systematic sensitivity analysis. Ultimately, the application of DML—and its extension to instrumental variables (DML-IV)—confirms and even strengthens the original findings of Alesina et al. (2013), proving that the cultural scar of the plough is a robust economic reality rather than a statistical artifact.

2 Original Theoretical Framework

2.1 Baseline Model

In their foundational theoretical framework, Alesina et al. (2013) model contemporary gender norms as a function of historical agricultural technologies. The baseline Ordinary Least Squares (OLS) regression takes the following form:

$$y_c = \alpha + \beta \cdot \text{plough use}_c + X_c^H \Gamma + X_c^C \Pi + \epsilon_c$$

Where y_c represents the contemporary gender role outcomes for country c (primarily Female Labor Force Participation in the year 2000). Our main variable of interest, plough use$_c$, estimates the proportion of a country's current citizens whose ancestors utilized the plough in pre-industrial agriculture.

To isolate the effect of the plough, the model explicitly divides potential confounders into two distinct vectors. X_c^H represents the historical controls, which include ethnographic variables such as the ancestral domestication of large animals and ancestral settlement patterns. X_c^C represents the contemporary controls, most notably the natural log of real per capita GDP and its square, which is included to accurately capture the known U-shaped relationship between economic development and female labor force participation[1]. The coefficient of interest is β, which captures the historical, long-term effect of the plough on modern gender roles.

2.2 Endogeneity Problem

Estimating the baseline model using standard OLS presents significant endogeneity challenges that compromise the causal interpretation of the estimates. The first major challenge is reverse causality. It is highly plausible that plough agriculture became more common in societies that already held less equal gender-role attitudes. Because preexisting cultural norms could drive the adoption of the technology, this reverse causality tends to bias the OLS estimates away from zero. The second major challenge is omitted variable bias. The historical adoption of the plough is strongly correlated with higher levels of early economic and societal development. Since higher development generally increases female labor force participation, failing to account for this factor effectively masks the plough's underlying negative impact on gender equality. Consequently, omitting this development factor biases the OLS estimates towards zero.

To resolve these issues, Alesina et al. (2013) propose two primary solutions. The first solution is to directly account for potential confounders by adding extensive

[1] This U-shaped relationship was famously theorized and empirically demonstrated by Goldin (1995), who showed that female labor force participation tends to initially decline during early industrialization before rising again as a country transitions into a high-income, service-based economy.

observable variables to the model. These observables are divided into two distinct groups: the historical characteristics of the modern population's ancestors, and the current economic, social, and cultural characteristics of the country.

However, because controlling for observables cannot fully eliminate the threat of unmeasured historical factors, the second solution relies on an Instrumental Variable (IV) strategy. This approach exploits the exogenous geographical fact that the historical adoption of the plough is strongly correlated with natural land suitability for specific crops. The researchers utilize two specific instruments: the land's suitability for "plough-positive" crops, such as wheat and barley, and its suitability for "plough-negative" crops, such as maize and sorghum[2]. The validity of this empirical strategy hinges on a strict exclusion restriction. It assumes that the ancestral suitability of the land for these specific crops affects contemporary gender norms solely through its historical influence on the adoption of the plough, rather than through any other direct economic or cultural channel.

3 Methodology

3.1 Why Causal Machine Learning?

The transition from traditional econometrics to Causal Machine Learning (CML) is driven by three primary methodological limitations inherent in the standard OLS approach. First, the researchers face the challenge of unknown functional forms. Economic and anthropological theory offers limited guidance on exactly how complex ancestral variables interact with contemporary cultural norms. Unlike traditional OLS, which requires researchers to manually guess and specify these relationships, CML algorithms flexibly and automatically capture these non-linear nuisance functions.

Second, attempting to manually account for all these complex interactions introduces a "curse of dimensionality." In their original study, Alesina et al. (2013) were constrained by a relatively small cross-country sample size ($N \approx 160$). If they were to include all potential raw covariates and their myriad interactions (p), the number of variables would quickly exceed the number of observations ($p > n$), rendering standard OLS estimation mathematically impossible due to a lack of degrees of freedom. CML handles this high-dimensional data environment through regularization. By systematically penalizing and shrinking the coefficients of less important variables, regularization allows for the simultaneous inclusion of all raw covariates without breaking the model.

Finally, the use of CML introduces a new standard of systematic robustness to empirical research. In traditional settings, the manual selection of control variables and functional forms to mitigate omitted variable bias is fundamentally an ad-hoc process. By implementing a CML framework, this ad-hoc variable selection is replaced with systematic, data-driven algorithms which objectively identify and select the most relevant confounders.

[2] "Plough-positive" crops require deep soil preparation and heavily benefit from the use of the plough, whereas "plough-negative" crops thrive in shallow soil and can be effectively cultivated using simple hand tools like hoes.

3.2 Double Machine Learning (DML)

To estimate the causal effect of historical plough use, Baiardi and Naghi (2024) employ a Partially Linear Regression (PLR) model:

$$
\begin{aligned}
Y &= D\theta_0 + g_0(X) + U, \quad \mathbb{E}[U|X, D] = 0 \\
D &= m_0(X) + V, \quad \mathbb{E}[V|X] = 0
\end{aligned}
$$

Y represents the outcome (Female Labor Force Participation), D is the treatment variable (historical plough use), and X is the high-dimensional set of controls. The functions $g_0(X)$ and $m_0(X)$ are unknown nuisance parameters representing the complex relationships between the controls, the outcome, and the treatment.

To estimate the Average Treatment Effect (ATE), denoted as θ_0, the DML framework utilizes a three-step "partialling out" estimation strategy. First, machine learning algorithms predict the outcome Y and the treatment D based purely on the covariates X. Second, these predictions are subtracted from the actual data to compute the respective residuals: $\tilde{Y} = Y - \mathbb{E}[Y|X]$ and $\tilde{D} = D - \mathbb{E}[D|X]$. Finally, the outcome residuals $\tilde{Y}$ are regressed on the treatment residuals $\tilde{D}$ to obtain θ_0. Intuitively, this strategy systematically removes the confounding effect of X from both the outcome and the treatment. By analyzing only the residuals, the model successfully isolates the "clean" variation in plough use, ensuring an unbiased causal estimate.

3.3 Double Machine Learning with IV (DML-IV)

While the standard DML model powerfully handles observed confounders, it relies on the assumption that all confounders are observed within the covariate matrix X. To account for *unobserved* confounders, the authors extend the framework to the Partially Linear IV (PLIV) model by incorporating two IVs, $Z_k = (Z_1, Z_2)$, which represent ancestral crop suitability. The model is defined by the following system of equations:

$$
\begin{aligned}
Y &= D\theta_0 + g_0(X) + \zeta, \quad \mathbb{E}[\zeta|Z, X] = 0 \\
D &= r_0(X) + T, \quad \mathbb{E}[T|X] = 0 \\
Z_k &= m_k(X) + V_k, \quad \mathbb{E}[V_k|X] = 0 \quad \text{for } k = 1, 2
\end{aligned}
$$

To estimate the true causal parameter θ_0, the estimation strategy expands into a three-step partialling-out process. First, in the prediction step, machine learning algorithms are utilized to predict the outcome Y, the treatment D, and the instruments Z using only the high-dimensional controls X. Second, in the residualization step, these ML predictions are subtracted from the actual data to compute the residuals for all three components ($\tilde{Y} = Y - \hat{Y}$, $\tilde{D} = D - \hat{D}$, and $\tilde{Z} = Z - \hat{Z}$). Finally, in the estimation step, θ_0 is isolated by performing Two-Stage Least Squares (TSLS) directly on these residuals. Specifically, this involves regressing the outcome residuals $\tilde{Y}$ on the treatment residuals $\tilde{D}$ while utilizing the residualized matrix $\tilde{Z}$ as the instruments.

4 Implementation

The code for the replication package is available at our GitHub repository. In the folder `/scripts`, the file `master.R` reproduces Tables A1 and A2 discussed in this report.

The implementation follows the workflow illustrated in Figure 1.

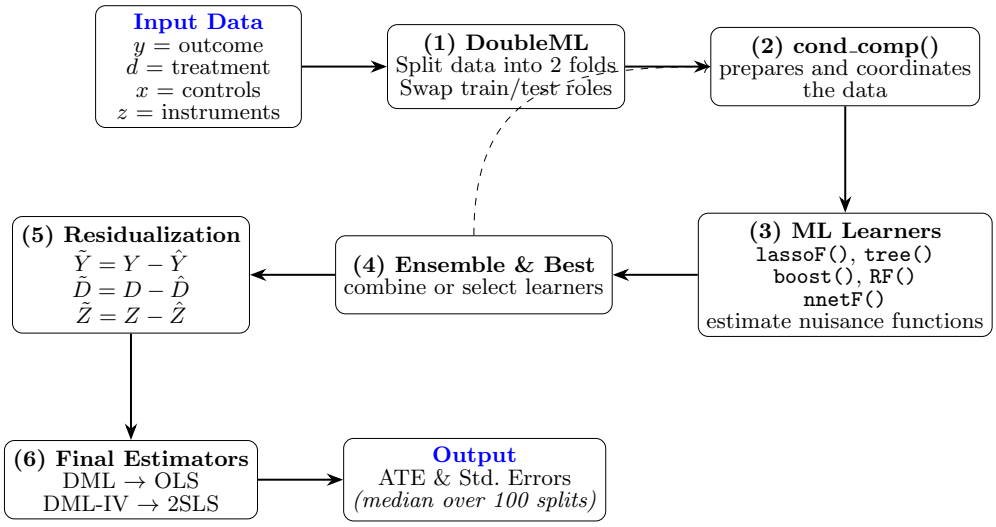

Fig. 1 Implementation pipeline of the customized Double Machine Learning procedure.

4.1 Software and Algorithms

We replicate the implementation of Baiardi and Naghi (2024) using a customized Double Machine Learning workflow in R. While the methodology is conceptually related to the `DoubleML` framework, the replication code relies on user-defined functions (such as `DoubleML()`, `cond_comp()`, and `ensembleF()`) to estimate nuisance functions, perform residualization, and compute the final causal parameters.

Although the official DoubleML package supports multiple instrumental variables within the PLIV framework, the authors' implementation introduces additional complexity. In particular, it combines ensemble predictions across heterogeneous learners while using different feature spaces: an expanded set of covariates (including nonlinearities and interactions) for Lasso, and raw covariates for the remaining algorithms.

The nuisance functions are estimated using a diverse set of learners. These include linear methods such as Lasso, which perform variable selection in high-dimensional settings; tree-based methods such as Regression Trees, Random Forests, and Boosting,

which flexibly capture nonlinearities and interactions; and Neural Networks, which provide an additional nonlinear approximation.

To aggregate predictions, the methodology uses two strategies. The **Best** approach selects the learner with the lowest out-of-sample mean squared error, while the **Ensemble** approach combines multiple learners through a weighted average.

4.2 Bias Correction via Cross-Fitting

A key component of the DML and DML-IV replication is *cross-fitting*, which is used to reduce overfitting bias when estimating nuisance functions with machine learning methods. For the pipeline presented in 1, the data is randomly partitioned into two folds. For each split, the nuisance functions are estimated on one fold and used to generate out-of-sample predictions on the other fold. The roles of the folds are then swapped, so that every observation receives an out-of-sample prediction. This procedure is essential because it ensures that the residualized variables are constructed using predictions obtained from models trained on separate data, thereby preserving valid inference (Chernozhukov et al., 2018).

In the replication code, this two-fold cross-fitting procedure is repeated **100 times** using different random sample partitions. Importantly, these 100 splits do not involve subsampling the data; the full sample is always used. What changes across iterations is only the random assignment of observations to folds. The final treatment effect is then reported as the median across the 100 estimates, and standard errors are adjusted using the median method.

4.3 Limitations

The function required to construct ensemble predictions for the instrument-related nuisance components is not available in the replication package. We suspect this limitation is closely related to the structure of the implementation. The Ensemble procedure requires combining predictions from learners that are trained on different feature spaces: an expanded set of covariates for Lasso and raw covariates for the remaining methods. Extending this dual-feature design to the IV setting requires simultaneously generating and aggregating predictions for the instrument-related nuisance functions, which is not supported by the available code. As a result, while individual learners could be successfully reproduced, the Ensemble-IV results reported in Table A2 Panel B could not be replicated.

5 Results

5.1 Replicating the Baseline

To verify that the original findings were not merely artifacts of functional form misspecification, the analysis first replicates the baseline linear model using Double Machine Learning (DML). As shown in Table A1, the DML partially linear model uses the original, limited set of seven covariates. The DML estimates for the treatment effect of the plough are negative, statistically significant, and quantitatively very close

to the original OLS estimates across all machine learning algorithms. This confirms that for a small set of covariates, the original assumption of linearity was adequate.

5.2 The Full Specification

The true advantage of CML emerges when testing the robustness of the baseline results against a high-dimensional set of controls. Panel A of Table A2 reports the DML estimates when expanding the covariates to 36 variables, including continent fixed effects and complex historical controls.

Under this rigorous stress test, the estimated coefficient for plough use remains negative across all learners but becomes substantially smaller in absolute magnitude (shrinking to approximately -6.0) and loses statistical significance in several specifications. This reduction indicates that traditional OLS likely overestimates the effect when failing to flexibly account for confounding observable characteristics, such as specific geo-climatic or historical development factors.

5.3 DML-IV Estimates

Having validated the instrument, the final step employs the DML-IV model to correct for unobserved confounders and measurement error. Panel B of Table A2 presents these results.

When accounting for hidden bias, the estimated coefficients are negative, generally statistically significant, and remarkably larger in magnitude (ranging from -21.8 to -41.7) than the baseline PLR estimates. This substantial increase suggests that the standard estimates are biased towards zero—likely due to attenuation bias from noisy historical data or unobserved traits that favored both plough adoption and female participation. Consequently, the DML-IV results provide compelling evidence that the historical "scar" of the plough is even deeper than traditional methods suggest.

6 Discussion and Key Takeaways

The application of CML by Baiardi and Naghi (2024) provides a robust empirical confirmation of the foundational findings by Alesina et al. (2013), demonstrating a significant negative causal effect of historical plough use on contemporary female labor force participation. Notably, the findings reveal that the DML-IV estimates are actually larger in magnitude than the original 2SLS results. While the authors do not formally test whether the difference between these two estimates is statistically significant, presenting them side-by-side illustrates a crucial methodological point: by utilizing flexible machine learning algorithms to systematically strip away high-dimensional confounding noise, the model successfully isolates a much stronger and cleaner causal signal.

Beyond these specific empirical results, this framework offers two crucial methodological takeaways. First, it introduces a new standard of systematic robustness to empirical economics. Rather than relying on the ad-hoc, manual variable selection typical of traditional robustness checks, CML employs data-driven algorithms to automatically incorporate the complex nonlinearities and interactions that would otherwise

exhaust standard OLS. Second, it highlights the absolute necessity of combining machine learning with IVs. While ML perfectly handles observed high-dimensional confounders, it cannot account for unobserved endogeneity—as evidenced by the fact that standard DML estimates without an instrument actually lost statistical significance. By integrating CML with an IV strategy, researchers can simultaneously overcome both observable and unobservable biases, providing a profoundly powerful toolkit for applied economists.

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Appendix A Tables and Figures

TABLE 1 Country-level estimates, partially linear model.

	(1) Lasso	(2) Reg. tree	(3) Boosting	(4) Forest	(5) Neural net.	(6) Ensemble	(7) Best	(8) OLS
	Panel A: Female labor force participation 2000							
Plough use	−10.533	−9.715	−11.934	−11.325	−11.663	−11.611	−11.404	−12.401
	(3.167)	(2.818)	(3.120	(3.178)	(3.177)	(3.179)	(3.192)	(2.964)
Observations	165	165	165	165	165	165	165	165
	Panel B: Share of firms with female ownership							
Plough use	−12.832	−11.543	−12.813	−12.566	−13.438	−13.337	−13.281	−15.241
	(4.006)	(3.680)	(3.959)	(4.482)	(3.873)	(4.172)	(4.285)	(4.06)
Observations	123	123	123	123	123	123	123	123
	Panel C: Share of political positions held by women 2000							
Plough use	−5.165	−3.670	−4.295	−5.287	−4.465	−5.072	−4.971	−4.821
	(1.957)	(1.764)	(1.786)	(1.803)	(1.901)	(1.901)	(1.858)	(1.782)
Observations	144	144	144	144	144	144	144	144
Raw covariates	7	7	7	7	7	7	7	7

Note: Analysis of Table 4 (columns 1, 3, and 5) of Alesina et al. (2013) using DML. Column 8 reports the original paper results. Standard errors adjusted for variability across splits using the median method are reported for the DML estimates. Robust standard errors are reported in column 8. The number of covariates does not include the treatment variable. The regressions include the baseline historical ethnographic and contemporary controls.
Abbreviation: OLS, ordinary least squares.

Fig. A1 Country-level estimates, partially linear model (Baseline replication with limited controls). Source: Baiardi & Naghi (2024).

TABLE 2 Country-level estimates with full set of controls.

	(1) Lasso	(2) Reg. tree	(3) Boosting	(4) Forest	(5) Neural net.	(6) Ensemble	(7) Best	(8) OLS
	Panel A: DML, Partially linear model. Outcome: Female labor force participation							
Plough use	−5.865	−5.951	−6.337	−6.023	−5.580	−6.062	−6.197	−9.234
	(4.892)	(4.056)	(4.235)	(4.265)	(4.740)	(4.433)	(4.416)	(4.301)
Observations	142	142	142	142	142	142	142	142
Raw covariates	36	36	36	36	36	36	36	30
	Lasso	**Reg. tree**	**Boosting**	**Forest**	**Neural net.**	**Ensemble**	**Best**	**2SLS**
	Panel B: DML-IV. Outcome: Female labor force participation							
Plough use	−41.742	−37.035	−38.766	−35.563	−21.882	−37.398	−35.898	−28.516
	(15.370)	(12.603)	(12.944)	(12.475)	(18.460)	(16.640)	(19.292)	(7.559)
Observations	160	160	160	160	160	160	160	160
Raw covariates	17	17	17	17	17	17	17	11

Note: Analysis of the main robustness checks of Alesina et al. (2013) using DML. Column 8 reports the results of the most complete robustness checks for the OLS and IV specifications in the original paper. Standard errors adjusted for variability across splits using the median method are reported for the DML estimates. Robust standard errors are reported in column 8. The number of covariates does not include the treatment variable. The regressions in panel A include the baseline set of controls, country fixed effects, plus additional historical controls and current economic, social, and cultural characteristics of countries. The regressions in panel B include the baseline set of controls, country fixed effects, and a number of geo-climatic characteristics.

Fig. A2 Country-level estimates. Panel A reports DML estimates with the full set of controls (36 covariates). Panel B reports DML-IV estimates addressing unobserved confounders. Source: Baiardi & Naghi (2024).

TABLE 3 First-stage estimates.

	(1) Lasso	(2) Reg. tree	(3) Boosting	(4) Forest	(5) Neural net.	(6) Ensemble	(7) Best	(8) OLS
Plough-positive crops	0.526	0.480	0.590	0.600	0.384	0.498	0.408	0.678
	(0.164)	(0.136)	(0.151)	(0.140)	(0.153)	(0.159)	(0.157)	(0.204)
Plough-negative crops	0.161	-0.065	0.142	0.087	0.226	0.221	0.152	0.232
	(0.158)	(0.149)	(0.153)	(0.157)	(0.129)	(0.141)	(0.146)	(0.204)
Observations	160	160	160	160	160	160	160	160

Note: This table reports the results of the first-stage analysis of Table 2, panel B. For the DML analysis, the standard errors are adjusted for variability across splits using the median method. Robust standard errors are reported in column 8.
Abbreviation: OLS, ordinary least squares.

Fig. A3 First-stage estimates predicting historical plough use based on crop suitability. Source: Baiardi & Naghi (2024).